Multi-Area OSPF IPv6

By: Grace Hwang

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Purpose

Building a multi-area OSPF with IPv6 for multiple networks in Cisco Packet Tracer helps facilitate a deeper understanding of OSPF, building off my experience with single-area OSPF. While my single-area OSPF lab helped to refresh my memory of general configurations, this lab helped me remember how to use IPv6 and recall how to connect networks over different areas. This lab deals a lot with being able to identify and diagnose any problems that may occur by pinging each device in the network to figure out if the devices between the two areas can connect or contact each other. After identifying the issue and where it occurs, the next step is to problem solve. Being able to discover issues and solve problems is a vital part of not just configuring networks, but it also applies to almost every other aspect of life.

Background Information

Multi-area OSPF v3 is used when you want the fastest or most efficient routing. OSPF allows for smaller routing tables, reduced link-state update (OSPF packet type 4: carries advertisements one hop further from its origin) overhead, and reduced frequency of SPF (shortest path tree) calculations. There are 5 important packet types in OSPFv3: Hello packet, Database Description (DD) packet, Link State Request (LSR) packet, Link State Update (LSU), and the Link State Acknowledgement (LSAck) packet. Hello packets are commonly used and occasionally are sent on OSPFv3 to establish and maintain relationships with neighboring routers. The DD packet is used by devices to describe their own link state database so they can synchronize. The LSR Packet is sent after two devices exchange DD packets to request each other's LSAs. The LSU packet is used to transmit Link-State Advertisements requested by its neighbors or to flood its own updated LSAs. Lastly, LSAck packets are used to acknowledge the LSAs that are found in a LSU packet.

OSPF was first created in the late 1980s. The Internet Engineering Task Force (IEFT) needed an IP routing protocol that worked well for large networks. OSPF was first used as a standard by John Moy. Later Version 2 was made, improving OSPF. After many modifications and alterations, OSPF was able to support IPv6 in Version 3. Despite these changes, the overall foundation and fundamentals of OSPF's mechanisms and algorithms did not change. However, the packets and LSA formats were changed in OSPFv3 due to Ipv6 having larger IP addresses (128 bits rather than 64).

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Lab Summary

In this lab, I created nine networks in order to connect five Cisco 4321 routers with a NIM-2T added to them as well as five computers across two areas. I aimed to be able to communicate between all of the devices, especially devices in different areas. Through the G0/0/0 interface of the router, a network was created between each router to one of five different computers, establishing five of the networks. The remaining four networks were between the routers, using the S0/1/1 and S0/1/0 interfaces. In each device, I configured them with IPv6 IP addresses and set up OSPFv3. I put the first two routers in area 0, two routers in area 1, and the router in between the two areas, router 3, was split between the two areas. Interface G0/0/0 and S0/1/1 of router 3 are in area 0 along with router 1 and 2 as well as PC 1, 2, and 3 while interface S0/1/0 of router 3 is in area 1 along with router 4 and 5 and PC 4 and 5. All the routers' IPv6 addresses start with 2001. The first six networks use the letters from 'a-f' to create the different networks, while the last three networks use db8, dc8, and dd8 for the second part of the address. The next five parts of the IPv6 address just have zeros so I write it in as ::, and finally the last section of the IPv6 address is used to identify devices in the same network (the computer is identified with a 3 at the end and the router interfaces are identified with a 1 or 2 at the end).

Lab Commands

IPv6 OSPF # Area #: written in the interfaces that will be connected across routers, it enables OSPFv3 on that interface, setting it up to be used to connect all the devices across the networks and areas

IPv6 Router OSPF #: enables OSPFv3 on the router

IPv6 Unicast-Routing: enables/allows the use of IPv6 on the router

Router-ID: creates the identity or name tag of the router

Show IPv6 OSPF Interface: shows information of all the interfaces that have OSPFv3 enabled

Show IPv6 OSPF Neighbor: displays the information of surrounding networks (or neighbors) that this network is next to

Show IPv6 Route: presents the routing tables with all the networks that the router can reach through OSPFv3

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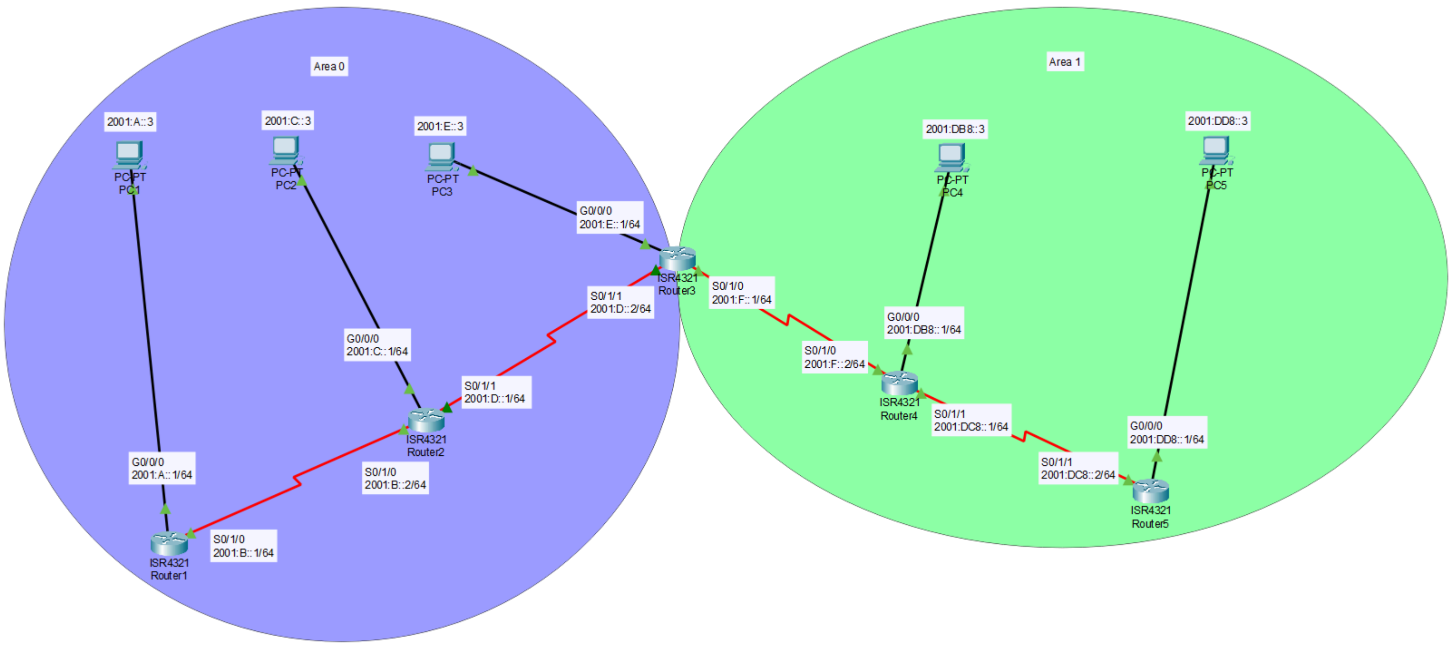
Network Diagram



Configurations

- **Router 1**

Show Run

```
ipv6 unicast-routing

interface GigabitEthernet0/0/0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:A::1/64
 ipv6 ospf 10 area 0

interface Serial0/1/0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:B::1/64
 ipv6 ospf 10 area 0

ipv6 router ospf 10
 router-id 1.1.1.1
 log-adjacency-changes
```

Show Ipv6 OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 1
```

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```
  Area 0, Process ID 10, Instance ID 0, Router ID 1.1.1.1
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, local address FE80::1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:04
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 3
  Area 0, Process ID 10, Instance ID 0, Router ID 1.1.1.1
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:04
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
```

Show Ipv6 OSPF Neighbor

```
Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
2.2.2.2           0   FULL/  -        00:00:38    3
Serial0/1/0
```

Show Ipv6 Route

```
C   2001:A::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:A::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:B::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:B::1/128 [0/0]
     via Serial0/1/0, receive
```

- 2001:C::/64 [110/65]

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```
     via FE80::1, Serial0/1/0
```

- 2001:D::/64 [110/128]
  via FE80::1, Serial0/1/0
- 2001:E::/64 [110/129]
  via FE80::1, Serial0/1/0

```
OI   2001:F::/64 [110/192]
     via FE80::1, Serial0/1/0
OI   2001:DB8::/64 [110/193]
     via FE80::1, Serial0/1/0
OI   2001:DC8::/64 [110/256]
     via FE80::1, Serial0/1/0
OI   2001:DD8::/64 [110/257]
     via FE80::1, Serial0/1/0
L    FF00::/8 [0/0]
     via Null0, receive
```

- **Router 2**

Show Run

```
ipv6 unicast-routing

interface GigabitEthernet0/0/0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:C::1/64
 ipv6 ospf 10 area 0

interface Serial0/1/0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:B::2/64
 ipv6 ospf 10 area 0
 clock rate 2000000

interface Serial0/1/1
 ipv6 address FE80::1 link-local
 ipv6 address 2001:D::1/64
 ipv6 ospf 10 area 0

ipv6 router ospf 10
 router-id 2.2.2.2
 log-adjacency-changes
```

Show Ipv6 OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 1
```

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```
  Area 0, Process ID 10, Instance ID 0, Router ID 2.2.2.2
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.2.2.2, local address FE80::1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:02
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 4
  Area 0, Process ID 10, Instance ID 0, Router ID 2.2.2.2
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:02
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 3
  Area 0, Process ID 10, Instance ID 0, Router ID 2.2.2.2
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:09
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)
```

Show Ipv6 OSPF Neighbor

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```
Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
1.1.1.1           0   FULL/  -        00:00:31    3
Serial0/1/0
3.3.3.3           0   FULL/  -        00:00:30    4
Serial0/1/1
```

Show Ipv6 Route

```
   ● 2001:A::/64 [110/65]
     via FE80::1, Serial0/1/0
C   2001:B::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:B::2/128 [0/0]
     via Serial0/1/0, receive
C   2001:C::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:C::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:D::/64 [0/0]
     via Serial0/1/1, directly connected
L   2001:D::1/128 [0/0]
     via Serial0/1/1, receive
   ● 2001:E::/64 [110/65]
     via FE80::1, Serial0/1/1
OI  2001:F::/64 [110/128]
     via FE80::1, Serial0/1/1
OI  2001:DB8::/64 [110/129]
     via FE80::1, Serial0/1/1
OI  2001:DC8::/64 [110/192]
     via FE80::1, Serial0/1/1
OI  2001:DD8::/64 [110/193]
     via FE80::1, Serial0/1/1
L   FF00::/8 [0/0]
     via Null0, receive
```

- **Router 3**

Show Run

```
ipv6 unicast-routing

interface GigabitEthernet0/0/0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:E::1/64
 ipv6 ospf 10 area 0
```

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```
interface Serial0/1/0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:F::1/64
 ipv6 ospf 10 area 1

interface Serial0/1/1
 ipv6 address FE80::1 link-local
 ipv6 address 2001:D::2/64
 ipv6 ospf 10 area 0
 clock rate 56000

ipv6 router ospf 10
 router-id 3.3.3.3
 log-adjacency-changes
```

Show Ipv6 OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 1
  Area 0, Process ID 10, Instance ID 0, Router ID 3.3.3.3
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, local address FE80::1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:04
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 4
  Area 0, Process ID 10, Instance ID 0, Router ID 3.3.3.3
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:04
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
```

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```
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 3
  Area 1, Process ID 10, Instance ID 0, Router ID 3.3.3.3
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:04
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 4.4.4.4
  Suppress hello for 0 neighbor(s)
```

Show Ipv6 OSPF Neighbor

```
Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
2.2.2.2           0   FULL/  -        00:00:30    4
Serial0/1/1
4.4.4.4           0   FULL/  -        00:00:38    3
Serial0/1/0
```

Show Ipv6 Route

```
  ●  2001:A::/64 [110/129]
     via FE80::1, Serial0/1/1
  ●  2001:B::/64 [110/128]
     via FE80::1, Serial0/1/1
  ●  2001:C::/64 [110/65]
     via FE80::1, Serial0/1/1
C   2001:D::/64 [0/0]
     via Serial0/1/1, directly connected
L   2001:D::2/128 [0/0]
     via Serial0/1/1, receive
C   2001:E::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:E::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:F::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:F::1/128 [0/0]
     via Serial0/1/0, receive
```

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- 2001:DB8::/64 [110/65]
  via FE80::1, Serial0/1/0
- 2001:DC8::/64 [110/128]
  via FE80::1, Serial0/1/0
- 2001:DD8::/64 [110/129]
  via FE80::1, Serial0/1/0

```
L   FF00::/8 [0/0]
     via Null0, receive
```

- **Router 4**

Show Run

```
ipv6 unicast-routing

interface GigabitEthernet0/0/0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8::1/64
 ipv6 ospf 10 area 1

interface Serial0/1/0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:F::2/64
 ipv6 ospf 10 area 1
 clock rate 2000000

interface Serial0/1/1
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DC8::1/64
 ipv6 ospf 10 area 1

ipv6 router ospf 10
 router-id 4.4.4.4
 log-adjacency-changes
```

Show Ipv6 OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 1
  Area 1, Process ID 10, Instance ID 0, Router ID 4.4.4.4
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 4.4.4.4, local address FE80::1
  No backup designated router on this network
```

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```
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:01
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 4
  Area 1, Process ID 10, Instance ID 0, Router ID 4.4.4.4
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:01
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 3
  Area 1, Process ID 10, Instance ID 0, Router ID 4.4.4.4
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:00
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

Show Ipv6 OSPF Neighbor

```
Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
3.3.3.3           0   FULL/  -        00:00:37    3
Serial0/1/0
```

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```
5.5.5.5           0   FULL/  -        00:00:37    4
Serial0/1/1
```

Show Ipv6 Route

```
OI  2001:A::/64 [110/193]
     via FE80::1, Serial0/1/0
OI  2001:B::/64 [110/192]
     via FE80::1, Serial0/1/0
OI  2001:C::/64 [110/129]
     via FE80::1, Serial0/1/0
OI  2001:D::/64 [110/128]
     via FE80::1, Serial0/1/0
OI  2001:E::/64 [110/65]
     via FE80::1, Serial0/1/0
C   2001:F::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:F::2/128 [0/0]
     via Serial0/1/0, receive
C   2001:DB8::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DC8::/64 [0/0]
     via Serial0/1/1, directly connected
L   2001:DC8::1/128 [0/0]
     via Serial0/1/1, receive
```

- 2001:DD8::/64 [110/65]
  via FE80::1, Serial0/1/1

```
L   FF00::/8 [0/0]
     via Null0, receive
```

- **Router 5**

Show Run

```
ipv6 unicast-routing

interface GigabitEthernet0/0/0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DD8::1/64
 ipv6 ospf 10 area 1

interface Serial0/1/1
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DC8::2/64
```

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```
 ipv6 ospf 10 area 1
 clock rate 2000000

ipv6 router ospf 10
 router-id 5.5.5.5
 log-adjacency-changes
```

Show Ipv6 OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 1
  Area 1, Process ID 10, Instance ID 0, Router ID 5.5.5.5
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 5.5.5.5, local address FE80::1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:01
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Link Local Address FE80::1, Interface ID 4
  Area 1, Process ID 10, Instance ID 0, Router ID 5.5.5.5
  Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:09
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 4.4.4.4
  Suppress hello for 0 neighbor(s)
```

Show Ipv6 OSPF Neighbor

```
Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
4.4.4.4           0   FULL/  -        00:00:39    4
Serial0/1/1
```

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Show Ipv6 Route

```
OI  2001:A::/64 [110/257]
     via FE80::1, Serial0/1/1
OI  2001:B::/64 [110/256]
     via FE80::1, Serial0/1/1
OI  2001:C::/64 [110/193]
     via FE80::1, Serial0/1/1
OI  2001:D::/64 [110/192]
     via FE80::1, Serial0/1/1
OI  2001:E::/64 [110/129]
     via FE80::1, Serial0/1/1
   ● 2001:F::/64 [110/128]
     via FE80::1, Serial0/1/1
   ● 2001:DB8::/64 [110/65]
     via FE80::1, Serial0/1/1
C   2001:DC8::/64 [0/0]
     via Serial0/1/1, directly connected
L   2001:DC8::2/128 [0/0]
     via Serial0/1/1, receive
C   2001:DD8::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DD8::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

- **PC1**

Ping R1

```
C:\>ping 2001:a::1

Pinging 2001:a::1 with 32 bytes of data:

Reply from 2001:A::1: bytes=32 time=56ms TTL=255
Reply from 2001:A::1: bytes=32 time<1ms TTL=255
Reply from 2001:A::1: bytes=32 time<1ms TTL=255
Reply from 2001:A::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 56ms, Average = 14ms

C:\>ping 2001:b::1
```

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```
Pinging 2001:b::1 with 32 bytes of data:

Reply from 2001:B::1: bytes=32 time<1ms TTL=255
Reply from 2001:B::1: bytes=32 time<1ms TTL=255
Reply from 2001:B::1: bytes=32 time<1ms TTL=255
Reply from 2001:B::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping R2

```
C:\>ping 2001:b::2

Pinging 2001:b::2 with 32 bytes of data:

Reply from 2001:B::2: bytes=32 time=55ms TTL=254
Reply from 2001:B::2: bytes=32 time=1ms TTL=254
Reply from 2001:B::2: bytes=32 time=1ms TTL=254
Reply from 2001:B::2: bytes=32 time=3ms TTL=254

Ping statistics for 2001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 55ms, Average = 15ms

C:\>ping 2001:c::1

Pinging 2001:c::1 with 32 bytes of data:

Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 2001:d::1

Pinging 2001:d::1 with 32 bytes of data:
```

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```
Reply from 2001:D::1: bytes=32 time=2ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:D::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping R3

```
C:\>ping 2001:d::2

Pinging 2001:d::2 with 32 bytes of data:

Reply from 2001:D::2: bytes=32 time=59ms TTL=253
Reply from 2001:D::2: bytes=32 time=2ms TTL=253
Reply from 2001:D::2: bytes=32 time=2ms TTL=253
Reply from 2001:D::2: bytes=32 time=11ms TTL=253

Ping statistics for 2001:D::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 59ms, Average = 18ms

C:\>ping 2001:e::1

Pinging 2001:e::1 with 32 bytes of data:

Reply from 2001:E::1: bytes=32 time=3ms TTL=253
Reply from 2001:E::1: bytes=32 time=2ms TTL=253
Reply from 2001:E::1: bytes=32 time=2ms TTL=253
Reply from 2001:E::1: bytes=32 time=4ms TTL=253

Ping statistics for 2001:E::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>ping 2001:f::1

Pinging 2001:f::1 with 32 bytes of data:

Reply from 2001:F::1: bytes=32 time=3ms TTL=253
Reply from 2001:F::1: bytes=32 time=4ms TTL=253
Reply from 2001:F::1: bytes=32 time=6ms TTL=253
```

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```
Reply from 2001:F::1: bytes=32 time=2ms TTL=253

Ping statistics for 2001:F::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms
```

Ping R4

```
C:\>ping 2001:f::2

Pinging 2001:f::2 with 32 bytes of data:

Reply from 2001:F::2: bytes=32 time=58ms TTL=252
Reply from 2001:F::2: bytes=32 time=5ms TTL=252
Reply from 2001:F::2: bytes=32 time=3ms TTL=252
Reply from 2001:F::2: bytes=32 time=3ms TTL=252

Ping statistics for 2001:F::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 58ms, Average = 17ms

C:\>ping 2001:db8::1

Pinging 2001:db8::1 with 32 bytes of data:

Reply from 2001:DB8::1: bytes=32 time=61ms TTL=252
Reply from 2001:DB8::1: bytes=32 time=14ms TTL=252
Reply from 2001:DB8::1: bytes=32 time=3ms TTL=252
Reply from 2001:DB8::1: bytes=32 time=3ms TTL=252

Ping statistics for 2001:DB8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 61ms, Average = 20ms

C:\>ping 2001:dc8::1

Pinging 2001:dc8::1 with 32 bytes of data:

Reply from 2001:DC8::1: bytes=32 time=5ms TTL=252
Reply from 2001:DC8::1: bytes=32 time=33ms TTL=252
Reply from 2001:DC8::1: bytes=32 time=10ms TTL=252
Reply from 2001:DC8::1: bytes=32 time=9ms TTL=252

Ping statistics for 2001:DC8::1:
```

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```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 33ms, Average = 14ms
```

Ping R5

```
C:\>ping 2001:dc8::2

Pinging 2001:dc8::2 with 32 bytes of data:

Reply from 2001:DC8::2: bytes=32 time=11ms TTL=251
Reply from 2001:DC8::2: bytes=32 time=11ms TTL=251
Reply from 2001:DC8::2: bytes=32 time=12ms TTL=251
Reply from 2001:DC8::2: bytes=32 time=8ms TTL=251

Ping statistics for 2001:DC8::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 12ms, Average = 10ms

C:\>ping 2001:dd8::1

Pinging 2001:dd8::1 with 32 bytes of data:

Reply from 2001:DD8::1: bytes=32 time=61ms TTL=251
Reply from 2001:DD8::1: bytes=32 time=12ms TTL=251
Reply from 2001:DD8::1: bytes=32 time=11ms TTL=251
Reply from 2001:DD8::1: bytes=32 time=11ms TTL=251

Ping statistics for 2001:DD8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 61ms, Average = 23ms
```

Ping PC2

```
C:\>ping 2001:c::3

Pinging 2001:c::3 with 32 bytes of data:

Reply from 2001:C::3: bytes=32 time=98ms TTL=126
Reply from 2001:C::3: bytes=32 time=2ms TTL=126
Reply from 2001:C::3: bytes=32 time=45ms TTL=126
Reply from 2001:C::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:C::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

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```
    Minimum = 1ms, Maximum = 98ms, Average = 36ms
```

Ping PC3

```
C:\>ping 2001:e::3

Pinging 2001:e::3 with 32 bytes of data:

Reply from 2001:E::3: bytes=32 time=62ms TTL=125
Reply from 2001:E::3: bytes=32 time=16ms TTL=125
Reply from 2001:E::3: bytes=32 time=25ms TTL=125
Reply from 2001:E::3: bytes=32 time=27ms TTL=125

Ping statistics for 2001:E::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 62ms, Average = 32ms
```

Ping PC4

```
C:\>ping 2001:db8::3

Pinging 2001:db8::3 with 32 bytes of data:

Reply from 2001:DB8::3: bytes=32 time=62ms TTL=124
Reply from 2001:DB8::3: bytes=32 time=3ms TTL=124
Reply from 2001:DB8::3: bytes=32 time=15ms TTL=124
Reply from 2001:DB8::3: bytes=32 time=7ms TTL=124

Ping statistics for 2001:DB8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 62ms, Average = 21ms
```

Ping PC5

```
C:\>ping 2001:dd8::3

Pinging 2001:dd8::3 with 32 bytes of data:

Reply from 2001:DD8::3: bytes=32 time=64ms TTL=123
Reply from 2001:DD8::3: bytes=32 time=13ms TTL=123
Reply from 2001:DD8::3: bytes=32 time=15ms TTL=123
Reply from 2001:DD8::3: bytes=32 time=12ms TTL=123

Ping statistics for 2001:DD8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

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```
    Minimum = 12ms, Maximum = 64ms, Average = 26ms
```

- **PC2**

Ping R1

```
C:\>ping 2001:a::1

Pinging 2001:a::1 with 32 bytes of data:

Reply from 2001:A::1: bytes=32 time=37ms TTL=254
Reply from 2001:A::1: bytes=32 time=2ms TTL=254
Reply from 2001:A::1: bytes=32 time=1ms TTL=254
Reply from 2001:A::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 37ms, Average = 10ms

C:\>ping 2001:b::1

Pinging 2001:b::1 with 32 bytes of data:

Reply from 2001:B::1: bytes=32 time=3ms TTL=254
Reply from 2001:B::1: bytes=32 time=1ms TTL=254
Reply from 2001:B::1: bytes=32 time=1ms TTL=254
Reply from 2001:B::1: bytes=32 time=10ms TTL=254

Ping statistics for 2001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

Ping R2

```
C:\>ping 2001:b::2

Pinging 2001:b::2 with 32 bytes of data:

Reply from 2001:B::2: bytes=32 time<1ms TTL=255
Reply from 2001:B::2: bytes=32 time<1ms TTL=255
Reply from 2001:B::2: bytes=32 time=1ms TTL=255
Reply from 2001:B::2: bytes=32 time<1ms TTL=255

Ping statistics for 2001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

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```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:c::1

Pinging 2001:c::1 with 32 bytes of data:

Reply from 2001:C::1: bytes=32 time<1ms TTL=255
Reply from 2001:C::1: bytes=32 time<1ms TTL=255
Reply from 2001:C::1: bytes=32 time<1ms TTL=255
Reply from 2001:C::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:d::1

Pinging 2001:d::1 with 32 bytes of data:

Reply from 2001:D::1: bytes=32 time<1ms TTL=255
Reply from 2001:D::1: bytes=32 time=1ms TTL=255
Reply from 2001:D::1: bytes=32 time<1ms TTL=255
Reply from 2001:D::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:D::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping R3

```
C:\>ping 2001:d::2

Pinging 2001:d::2 with 32 bytes of data:

Reply from 2001:D::2: bytes=32 time=5ms TTL=254
Reply from 2001:D::2: bytes=32 time=2ms TTL=254
Reply from 2001:D::2: bytes=32 time=1ms TTL=254
Reply from 2001:D::2: bytes=32 time=2ms TTL=254

Ping statistics for 2001:D::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 2001:e::1
```

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```
Pinging 2001:e::1 with 32 bytes of data:

Reply from 2001:E::1: bytes=32 time=2ms TTL=254
Reply from 2001:E::1: bytes=32 time=12ms TTL=254
Reply from 2001:E::1: bytes=32 time=1ms TTL=254
Reply from 2001:E::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:E::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 4ms

C:\>ping 2001:f::1

Pinging 2001:f::1 with 32 bytes of data:

Reply from 2001:F::1: bytes=32 time=2ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:F::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping R4

```
C:\>ping 2001:f::2

Pinging 2001:f::2 with 32 bytes of data:

Reply from 2001:F::2: bytes=32 time=3ms TTL=253
Reply from 2001:F::2: bytes=32 time=2ms TTL=253
Reply from 2001:F::2: bytes=32 time=2ms TTL=253
Reply from 2001:F::2: bytes=32 time=2ms TTL=253

Ping statistics for 2001:F::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 2001:db8::1

Pinging 2001:db8::1 with 32 bytes of data:
```

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```
Reply from 2001:DB8::1: bytes=32 time=59ms TTL=253
Reply from 2001:DB8::1: bytes=32 time=13ms TTL=253
Reply from 2001:DB8::1: bytes=32 time=2ms TTL=253
Reply from 2001:DB8::1: bytes=32 time=7ms TTL=253

Ping statistics for 2001:DB8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 59ms, Average = 20ms

C:\>ping 2001:dc8::1

Pinging 2001:dc8::1 with 32 bytes of data:

Reply from 2001:DC8::1: bytes=32 time=3ms TTL=253
Reply from 2001:DC8::1: bytes=32 time=2ms TTL=253
Reply from 2001:DC8::1: bytes=32 time=6ms TTL=253
Reply from 2001:DC8::1: bytes=32 time=14ms TTL=253

Ping statistics for 2001:DC8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 14ms, Average = 6ms
```

Ping R5

```
C:\>ping 2001:dc8::2

Pinging 2001:dc8::2 with 32 bytes of data:

Reply from 2001:DC8::2: bytes=32 time=6ms TTL=252
Reply from 2001:DC8::2: bytes=32 time=14ms TTL=252
Reply from 2001:DC8::2: bytes=32 time=5ms TTL=252
Reply from 2001:DC8::2: bytes=32 time=15ms TTL=252

Ping statistics for 2001:DC8::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 15ms, Average = 10ms

C:\>ping 2001:dd8::1

Pinging 2001:dd8::1 with 32 bytes of data:

Reply from 2001:DD8::1: bytes=32 time=4ms TTL=252
Reply from 2001:DD8::1: bytes=32 time=5ms TTL=252
Reply from 2001:DD8::1: bytes=32 time=5ms TTL=252
```

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```
Reply from 2001:DD8::1: bytes=32 time=12ms TTL=252

Ping statistics for 2001:DD8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 12ms, Average = 6ms
```

Ping PC1

```
C:\>ping 2001:a::3

Pinging 2001:a::3 with 32 bytes of data:

Reply from 2001:A::3: bytes=32 time=1ms TTL=126
Reply from 2001:A::3: bytes=32 time=5ms TTL=126
Reply from 2001:A::3: bytes=32 time=1ms TTL=126
Reply from 2001:A::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:A::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 2ms
```

Ping PC3

```
C:\>ping 2001:e::3

Pinging 2001:e::3 with 32 bytes of data:

Reply from 2001:E::3: bytes=32 time=1ms TTL=126
Reply from 2001:E::3: bytes=32 time=2ms TTL=126
Reply from 2001:E::3: bytes=32 time=1ms TTL=126
Reply from 2001:E::3: bytes=32 time=10ms TTL=126

Ping statistics for 2001:E::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

Ping PC4

```
C:\>ping 2001:db8::3

Pinging 2001:db8::3 with 32 bytes of data:

Reply from 2001:DB8::3: bytes=32 time=81ms TTL=125
Reply from 2001:DB8::3: bytes=32 time=2ms TTL=125
Reply from 2001:DB8::3: bytes=32 time=4ms TTL=125
```

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```
Reply from 2001:DB8::3: bytes=32 time=2ms TTL=125

Ping statistics for 2001:DB8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 81ms, Average = 22ms
```

Ping PC5

```
C:\>ping 2001:dd8::3

Pinging 2001:dd8::3 with 32 bytes of data:

Reply from 2001:DD8::3: bytes=32 time=54ms TTL=124
Reply from 2001:DD8::3: bytes=32 time=11ms TTL=124
Reply from 2001:DD8::3: bytes=32 time=11ms TTL=124
Reply from 2001:DD8::3: bytes=32 time=11ms TTL=124

Ping statistics for 2001:DD8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 54ms, Average = 21ms
```

- **PC3**

Ping R1

```
C:\>ping 2001:a::1

Pinging 2001:a::1 with 32 bytes of data:

Reply from 2001:A::1: bytes=32 time=3ms TTL=253
Reply from 2001:A::1: bytes=32 time=8ms TTL=253
Reply from 2001:A::1: bytes=32 time=2ms TTL=253
Reply from 2001:A::1: bytes=32 time=2ms TTL=253

Ping statistics for 2001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 8ms, Average = 3ms

C:\>ping 2001:b::1

Pinging 2001:b::1 with 32 bytes of data:

Reply from 2001:B::1: bytes=32 time=3ms TTL=253
Reply from 2001:B::1: bytes=32 time=11ms TTL=253
Reply from 2001:B::1: bytes=32 time=4ms TTL=253
```

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```
Reply from 2001:B::1: bytes=32 time=2ms TTL=253

Ping statistics for 2001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 5ms
```

Ping R2

```
C:\>ping 2001:b::2

Pinging 2001:b::2 with 32 bytes of data:

Reply from 2001:B::2: bytes=32 time=40ms TTL=254
Reply from 2001:B::2: bytes=32 time=3ms TTL=254
Reply from 2001:B::2: bytes=32 time=1ms TTL=254
Reply from 2001:B::2: bytes=32 time=3ms TTL=254

Ping statistics for 2001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 40ms, Average = 11ms

C:\>ping 2001:c::1

Pinging 2001:c::1 with 32 bytes of data:

Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254
Reply from 2001:C::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 2001:d::1

Pinging 2001:d::1 with 32 bytes of data:

Reply from 2001:D::1: bytes=32 time=1ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254
Reply from 2001:D::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:D::1:
```

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```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Ping R3

```
C:\>ping 2001:d::2

Pinging 2001:d::2 with 32 bytes of data:

Reply from 2001:D::2: bytes=32 time=1ms TTL=255
Reply from 2001:D::2: bytes=32 time<1ms TTL=255
Reply from 2001:D::2: bytes=32 time<1ms TTL=255
Reply from 2001:D::2: bytes=32 time=1ms TTL=255

Ping statistics for 2001:D::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:e::1

Pinging 2001:e::1 with 32 bytes of data:

Reply from 2001:E::1: bytes=32 time=1ms TTL=255
Reply from 2001:E::1: bytes=32 time<1ms TTL=255
Reply from 2001:E::1: bytes=32 time<1ms TTL=255
Reply from 2001:E::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:E::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:f::1

Pinging 2001:f::1 with 32 bytes of data:

Reply from 2001:F::1: bytes=32 time<1ms TTL=255
Reply from 2001:F::1: bytes=32 time<1ms TTL=255
Reply from 2001:F::1: bytes=32 time<1ms TTL=255
Reply from 2001:F::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:F::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

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Ping R4

```
C:\>ping 2001:f::2

Pinging 2001:f::2 with 32 bytes of data:

Reply from 2001:F::2: bytes=32 time=45ms TTL=254
Reply from 2001:F::2: bytes=32 time=3ms TTL=254
Reply from 2001:F::2: bytes=32 time=3ms TTL=254
Reply from 2001:F::2: bytes=32 time=3ms TTL=254

Ping statistics for 2001:F::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 45ms, Average = 13ms

C:\>ping 2001:db8::1

Pinging 2001:db8::1 with 32 bytes of data:

Reply from 2001:DB8::1: bytes=32 time=4ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=3ms TTL=254

Ping statistics for 2001:DB8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>ping 2001:dc8::1

Pinging 2001:dc8::1 with 32 bytes of data:

Reply from 2001:DC8::1: bytes=32 time=2ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=4ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=2ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=3ms TTL=254

Ping statistics for 2001:DC8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms
```

Ping R5

```
C:\>ping 2001:dc8::2
```

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```
Pinging 2001:dc8::2 with 32 bytes of data:

Reply from 2001:DC8::2: bytes=32 time=61ms TTL=253
Reply from 2001:DC8::2: bytes=32 time=2ms TTL=253
Reply from 2001:DC8::2: bytes=32 time=6ms TTL=253
Reply from 2001:DC8::2: bytes=32 time=2ms TTL=253

Ping statistics for 2001:DC8::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 61ms, Average = 17ms

C:\>ping 2001:dd8::1

Pinging 2001:dd8::1 with 32 bytes of data:

Reply from 2001:DD8::1: bytes=32 time=3ms TTL=253
Reply from 2001:DD8::1: bytes=32 time=4ms TTL=253
Reply from 2001:DD8::1: bytes=32 time=6ms TTL=253
Reply from 2001:DD8::1: bytes=32 time=12ms TTL=253

Ping statistics for 2001:DD8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 12ms, Average = 6ms
```

Ping PC1

```
C:\>ping 2001:a::3

Pinging 2001:a::3 with 32 bytes of data:

Reply from 2001:A::3: bytes=32 time=3ms TTL=125
Reply from 2001:A::3: bytes=32 time=13ms TTL=125
Reply from 2001:A::3: bytes=32 time=12ms TTL=125
Reply from 2001:A::3: bytes=32 time=6ms TTL=125

Ping statistics for 2001:A::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 13ms, Average = 8ms
```

Ping PC2

```
C:\>ping 2001:c::3

Pinging 2001:c::3 with 32 bytes of data:
```

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```
Reply from 2001:C::3: bytes=32 time=4ms TTL=126
Reply from 2001:C::3: bytes=32 time=1ms TTL=126
Reply from 2001:C::3: bytes=32 time=1ms TTL=126
Reply from 2001:C::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:C::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 1ms
```

Ping PC4

```
C:\>ping 2001:db8::3

Pinging 2001:db8::3 with 32 bytes of data:

Reply from 2001:DB8::3: bytes=32 time=1ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=2ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=1ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:DB8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping PC5

```
C:\>ping 2001:dd8::3

Pinging 2001:dd8::3 with 32 bytes of data:

Reply from 2001:DD8::3: bytes=32 time=4ms TTL=125
Reply from 2001:DD8::3: bytes=32 time=5ms TTL=125
Reply from 2001:DD8::3: bytes=32 time=12ms TTL=125
Reply from 2001:DD8::3: bytes=32 time=4ms TTL=125

Ping statistics for 2001:DD8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 12ms, Average = 6ms
```

- **PC4**

Ping R1

```
C:\>ping 2001:a::1
```

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```
Pinging 2001:a::1 with 32 bytes of data:

Reply from 2001:A::1: bytes=32 time=4ms TTL=252
Reply from 2001:A::1: bytes=32 time=3ms TTL=252
Reply from 2001:A::1: bytes=32 time=3ms TTL=252
Reply from 2001:A::1: bytes=32 time=3ms TTL=252

Ping statistics for 2001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>ping 2001:b::1

Pinging 2001:b::1 with 32 bytes of data:

Reply from 2001:B::1: bytes=32 time=4ms TTL=252
Reply from 2001:B::1: bytes=32 time=3ms TTL=252
Reply from 2001:B::1: bytes=32 time=5ms TTL=252
Reply from 2001:B::1: bytes=32 time=6ms TTL=252

Ping statistics for 2001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 4ms
```

Ping R2

```
C:\>ping 2001:b::2

Pinging 2001:b::2 with 32 bytes of data:

Reply from 2001:B::2: bytes=32 time=3ms TTL=253
Reply from 2001:B::2: bytes=32 time=12ms TTL=253
Reply from 2001:B::2: bytes=32 time=2ms TTL=253
Reply from 2001:B::2: bytes=32 time=4ms TTL=253

Ping statistics for 2001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 5ms

C:\>ping 2001:c::1

Pinging 2001:c::1 with 32 bytes of data:
```

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```
Reply from 2001:C::1: bytes=32 time=13ms TTL=253
Reply from 2001:C::1: bytes=32 time=2ms TTL=253
Reply from 2001:C::1: bytes=32 time=11ms TTL=253
Reply from 2001:C::1: bytes=32 time=2ms TTL=253

Ping statistics for 2001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 7ms

C:\>ping 2001:d::1

Pinging 2001:d::1 with 32 bytes of data:

Reply from 2001:D::1: bytes=32 time=3ms TTL=253
Reply from 2001:D::1: bytes=32 time=4ms TTL=253
Reply from 2001:D::1: bytes=32 time=2ms TTL=253
Reply from 2001:D::1: bytes=32 time=2ms TTL=253

Ping statistics for 2001:D::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 2ms
```

Ping R3

```
C:\>ping 2001:d::2

Pinging 2001:d::2 with 32 bytes of data:

Reply from 2001:D::2: bytes=32 time=2ms TTL=254
Reply from 2001:D::2: bytes=32 time=1ms TTL=254
Reply from 2001:D::2: bytes=32 time=1ms TTL=254
Reply from 2001:D::2: bytes=32 time=1ms TTL=254

Ping statistics for 2001:D::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 2001:e::1

Pinging 2001:e::1 with 32 bytes of data:

Reply from 2001:E::1: bytes=32 time=1ms TTL=254
Reply from 2001:E::1: bytes=32 time=5ms TTL=254
Reply from 2001:E::1: bytes=32 time=1ms TTL=254
```

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```
Reply from 2001:E::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:E::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 2001:f::1

Pinging 2001:f::1 with 32 bytes of data:

Reply from 2001:F::1: bytes=32 time=2ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254
Reply from 2001:F::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:F::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping R4

```
C:\>ping 2001:f::2

Pinging 2001:f::2 with 32 bytes of data:

Reply from 2001:F::2: bytes=32 time<1ms TTL=255
Reply from 2001:F::2: bytes=32 time<1ms TTL=255
Reply from 2001:F::2: bytes=32 time<1ms TTL=255
Reply from 2001:F::2: bytes=32 time<1ms TTL=255

Ping statistics for 2001:F::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8::1

Pinging 2001:db8::1 with 32 bytes of data:

Reply from 2001:DB8::1: bytes=32 time=8ms TTL=255
Reply from 2001:DB8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8::1: bytes=32 time=9ms TTL=255

Ping statistics for 2001:DB8::1:
```

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```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 4ms

C:\>ping 2001:dc8::1

Pinging 2001:dc8::1 with 32 bytes of data:

Reply from 2001:DC8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DC8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DC8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DC8::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DC8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping R5

```
C:\>ping 2001:dc8::2

Pinging 2001:dc8::2 with 32 bytes of data:

Reply from 2001:DC8::2: bytes=32 time=2ms TTL=254
Reply from 2001:DC8::2: bytes=32 time=1ms TTL=254
Reply from 2001:DC8::2: bytes=32 time=3ms TTL=254
Reply from 2001:DC8::2: bytes=32 time=4ms TTL=254

Ping statistics for 2001:DC8::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>ping 2001:dd8::1

Pinging 2001:dd8::1 with 32 bytes of data:

Reply from 2001:DD8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DD8::1: bytes=32 time=3ms TTL=254
Reply from 2001:DD8::1: bytes=32 time=3ms TTL=254
Reply from 2001:DD8::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DD8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

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Ping PC1

```
C:\>ping 2001:a::3

Pinging 2001:a::3 with 32 bytes of data:

Reply from 2001:A::3: bytes=32 time=87ms TTL=124
Reply from 2001:A::3: bytes=32 time=48ms TTL=124
Reply from 2001:A::3: bytes=32 time=12ms TTL=124
Reply from 2001:A::3: bytes=32 time=11ms TTL=124

Ping statistics for 2001:A::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 87ms, Average = 39ms
```

Ping PC2

```
C:\>ping 2001:c::3

Pinging 2001:c::3 with 32 bytes of data:

Reply from 2001:C::3: bytes=32 time=54ms TTL=125
Reply from 2001:C::3: bytes=32 time=13ms TTL=125
Reply from 2001:C::3: bytes=32 time=11ms TTL=125
Reply from 2001:C::3: bytes=32 time=4ms TTL=125

Ping statistics for 2001:C::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 54ms, Average = 20ms
```

Ping PC3

```
C:\>ping 2001:e::3

Pinging 2001:e::3 with 32 bytes of data:

Reply from 2001:E::3: bytes=32 time=50ms TTL=126
Reply from 2001:E::3: bytes=32 time=1ms TTL=126
Reply from 2001:E::3: bytes=32 time=3ms TTL=126
Reply from 2001:E::3: bytes=32 time=3ms TTL=126

Ping statistics for 2001:E::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

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```
    Minimum = 1ms, Maximum = 50ms, Average = 14ms
```

Ping PC5

```
C:\>ping 2001:dd8::3

Pinging 2001:dd8::3 with 32 bytes of data:

Reply from 2001:DD8::3: bytes=32 time=85ms TTL=126
Reply from 2001:DD8::3: bytes=32 time=11ms TTL=126
Reply from 2001:DD8::3: bytes=32 time=17ms TTL=126
Reply from 2001:DD8::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:DD8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 85ms, Average = 28ms
```

- **PC5**

Ping R1

```
C:\>ping 2001:a::1

Pinging 2001:a::1 with 32 bytes of data:

Reply from 2001:A::1: bytes=32 time=11ms TTL=251
Reply from 2001:A::1: bytes=32 time=4ms TTL=251
Reply from 2001:A::1: bytes=32 time=12ms TTL=251
Reply from 2001:A::1: bytes=32 time=14ms TTL=251

Ping statistics for 2001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 14ms, Average = 10ms

C:\>ping 2001:b::1

Pinging 2001:b::1 with 32 bytes of data:

Reply from 2001:B::1: bytes=32 time=5ms TTL=251
Reply from 2001:B::1: bytes=32 time=12ms TTL=251
Reply from 2001:B::1: bytes=32 time=11ms TTL=251
Reply from 2001:B::1: bytes=32 time=4ms TTL=251

Ping statistics for 2001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 12ms, Average = 8ms
```

Ping R2

```
C:\>ping 2001:b::2

Pinging 2001:b::2 with 32 bytes of data:

Reply from 2001:B::2: bytes=32 time=3ms TTL=252
Reply from 2001:B::2: bytes=32 time=6ms TTL=252
Reply from 2001:B::2: bytes=32 time=10ms TTL=252
Reply from 2001:B::2: bytes=32 time=5ms TTL=252

Ping statistics for 2001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 10ms, Average = 6ms

C:\>ping 2001:c::1

Pinging 2001:c::1 with 32 bytes of data:

Reply from 2001:C::1: bytes=32 time=12ms TTL=252
Reply from 2001:C::1: bytes=32 time=5ms TTL=252
Reply from 2001:C::1: bytes=32 time=6ms TTL=252
Reply from 2001:C::1: bytes=32 time=6ms TTL=252

Ping statistics for 2001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 12ms, Average = 7ms

C:\>ping 2001:d::1

Pinging 2001:d::1 with 32 bytes of data:

Reply from 2001:D::1: bytes=32 time=60ms TTL=252
Reply from 2001:D::1: bytes=32 time=12ms TTL=252
Reply from 2001:D::1: bytes=32 time=12ms TTL=252
Reply from 2001:D::1: bytes=32 time=5ms TTL=252

Ping statistics for 2001:D::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 60ms, Average = 22ms
```

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Ping R3

```
C:\>ping 2001:d::2

Pinging 2001:d::2 with 32 bytes of data:

Reply from 2001:D::2: bytes=32 time=44ms TTL=253
Reply from 2001:D::2: bytes=32 time=2ms TTL=253
Reply from 2001:D::2: bytes=32 time=3ms TTL=253
Reply from 2001:D::2: bytes=32 time=2ms TTL=253

Ping statistics for 2001:D::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 44ms, Average = 12ms

C:\>ping 2001:e::1

Pinging 2001:e::1 with 32 bytes of data:

Reply from 2001:E::1: bytes=32 time=3ms TTL=253
Reply from 2001:E::1: bytes=32 time=2ms TTL=253
Reply from 2001:E::1: bytes=32 time=12ms TTL=253
Reply from 2001:E::1: bytes=32 time=3ms TTL=253

Ping statistics for 2001:E::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 5ms

C:\>ping 2001:f::1

Pinging 2001:f::1 with 32 bytes of data:

Reply from 2001:F::1: bytes=32 time=3ms TTL=253
Reply from 2001:F::1: bytes=32 time=2ms TTL=253
Reply from 2001:F::1: bytes=32 time=2ms TTL=253
Reply from 2001:F::1: bytes=32 time=7ms TTL=253

Ping statistics for 2001:F::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms
```

Ping R4

```
C:\>ping 2001:f::2
```

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```
Pinging 2001:f::2 with 32 bytes of data:

Reply from 2001:F::2: bytes=32 time=2ms TTL=254
Reply from 2001:F::2: bytes=32 time=4ms TTL=254
Reply from 2001:F::2: bytes=32 time=1ms TTL=254
Reply from 2001:F::2: bytes=32 time=1ms TTL=254

Ping statistics for 2001:F::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>ping 2001:db8::1

Pinging 2001:db8::1 with 32 bytes of data:

Reply from 2001:DB8::1: bytes=32 time=4ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DB8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 1ms

C:\>ping 2001:dc8::1

Pinging 2001:dc8::1 with 32 bytes of data:

Reply from 2001:DC8::1: bytes=32 time=2ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=3ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=1ms TTL=254
Reply from 2001:DC8::1: bytes=32 time=5ms TTL=254

Ping statistics for 2001:DC8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 2ms
```

Ping R5

```
C:\>ping 2001:dc8::2

Pinging 2001:dc8::2 with 32 bytes of data:

Reply from 2001:DC8::2: bytes=32 time<1ms TTL=255
```

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```
Reply from 2001:DC8::2: bytes=32 time<1ms TTL=255
Reply from 2001:DC8::2: bytes=32 time<1ms TTL=255
Reply from 2001:DC8::2: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DC8::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:dd8::1

Pinging 2001:dd8::1 with 32 bytes of data:

Reply from 2001:DD8::1: bytes=32 time=1ms TTL=255
Reply from 2001:DD8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DD8::1: bytes=32 time<1ms TTL=255
Reply from 2001:DD8::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DD8::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping PC1

```
C:\>ping 2001:a::3

Pinging 2001:a::3 with 32 bytes of data:

Reply from 2001:A::3: bytes=32 time=14ms TTL=123
Reply from 2001:A::3: bytes=32 time=27ms TTL=123
Reply from 2001:A::3: bytes=32 time=12ms TTL=123
Reply from 2001:A::3: bytes=32 time=13ms TTL=123

Ping statistics for 2001:A::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 27ms, Average = 16ms
```

Ping PC2

```
C:\>ping 2001:c::3

Pinging 2001:c::3 with 32 bytes of data:

Reply from 2001:C::3: bytes=32 time=53ms TTL=124
Reply from 2001:C::3: bytes=32 time=11ms TTL=124
Reply from 2001:C::3: bytes=32 time=12ms TTL=124
```

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```
Reply from 2001:C::3: bytes=32 time=12ms TTL=124

Ping statistics for 2001:C::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 53ms, Average = 22ms
```

Ping PC3

```
C:\>ping 2001:e::3

Pinging 2001:e::3 with 32 bytes of data:

Reply from 2001:E::3: bytes=32 time=40ms TTL=125
Reply from 2001:E::3: bytes=32 time=2ms TTL=125
Reply from 2001:E::3: bytes=32 time=4ms TTL=125
Reply from 2001:E::3: bytes=32 time=2ms TTL=125

Ping statistics for 2001:E::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 40ms, Average = 12ms
```

Ping PC4

```
C:\>ping 2001:db8::3

Pinging 2001:db8::3 with 32 bytes of data:

Reply from 2001:DB8::3: bytes=32 time=3ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=3ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=1ms TTL=126
Reply from 2001:DB8::3: bytes=32 time=1ms TTL=126

Ping statistics for 2001:DB8::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

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Problems

One issue I encountered occurred because I started this lab by building off my single-area OSPF lab, where I had already implemented IPv6 addresses and router IDs. However, I had set different process IDs on each router. I learned that part of the reason my multi-area OSPF with IPv6 wasn't working was due to the use of different process IDs. From this mistake, I realized the importance of using the same process ID across all routers, so I corrected it by assigning the same process ID to each one.

Another challenge I faced occurred while altering the process IDs. An error message repeatedly appeared on my routers, indicating that I had mismatched process IDs. This made it difficult to work, as the message kept popping up and obscured what I was doing. I decided to revert to a previous version of my lab—before I had changed the process IDs—and discovered that I needed to remove the old process IDs before configuring the new ones. I removed the router IDs I had set on the old process IDs, disabled the processes, set the same process ID on all the routers in my topology, and then reconfigured the router ID on each router.

Finally, although all configurations appeared correct and I could ping PC5 in Area 1 from PC3 in Area 0 (which was connected to R3), I was still unable to ping R2, which was also in Area 0. After using commands like "show run", "show ipv6 ospf int," and others—and with the devices still unable to fully communicate—I decided to let time pass in Packet Tracer by speeding up the simulation. Eventually, I was able to ping all the routers and computers across both areas successfully.

Conclusion

Configuring a multi-area OSPF with IPv6 involving nine networks, five routers, and five computers—building off my previous single-area OSPF with IPv4—gave me valuable experience in modifying or fixing an existing setup, a situation I might encounter as a network engineer. Additionally, this lab served as a reminder of how to work with IPv6 addresses and highlighted the differences between configuring IPv6 and IPv4. It was also a helpful refresher on setting up multi-area OSPF and enabling communication between multiple areas.

This lab taught me the importance of patience; sometimes, you just have to wait for configurations to take effect before they work as expected. It also helped me strengthen my problem-solving skills, which are crucial for real-world roles in this field. Moreover, I gained more familiarity with Packet Tracer, including how to visually organize my topology by adding circles to represent different areas and using textboxes for labeling.

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